

1. Minimize the function $F(A, B, C) = A'B'C + A'BC + AB'C + ABC$ using Boolean laws.

- (A) C (B) $A + B$
(C) $A' + C$ (D) $B' + C$

2. Simplify the expression: $F = (A + B + C) \cdot (A + B' + C) \cdot (A + B + C')$.

- (A) $A + B \cdot C$ (B) $A \cdot B + C$
(C) $(A + B) \cdot (A + C)$ (D) $A + B + C$

3. How many Prime Implicants (PIs) are present in the function $F(A, B, C, D) = \sum m(0, 1, 2, 5, 8, 9, 10)$?

- (A) 3 (B) 2
(C) 4 (D) 5

4. Find the minimal SOP for $F(A, B, C, D) = \sum m(1, 3, 7, 11, 15) + d(0, 2, 5)$.

- (A) $C \cdot D + A'$ (B) $A \cdot D + B'$
(C) $C \cdot D + B \cdot D$ (D) $A' \cdot B + C'$

5. The Boolean function $F = X \cdot Y + X' \cdot Z + Y \cdot Z$ contains a redundant term. Which term is redundant?

- (A) $Y \cdot Z$ (B) $X \cdot Y$
(C) $X' \cdot Z$ (D) None

6. What is the output expression for an XNOR gate receiving inputs A' and B ?

- (A) $A \cdot B + A' \cdot B'$ (B) $A' \cdot B + A \cdot B'$
(C) $A \cdot B' + A' \cdot B$ (D) $(A + B)'$

7. Convert $F(A, B, C) = \prod M(0, 2, 3, 7)$ into standard Sum of Minterms form $\sum m$.

- (A) $\sum m(1, 4, 5, 6)$ (B) $\sum m(0, 2, 3, 7)$
(C) $\sum m(1, 2, 5, 6)$ (D) $\sum m(4, 5, 6, 7)$

8. Simplify the Boolean expression: $F = A \cdot B + A \cdot (B + C) + B \cdot (B + C)$.

- (A) $B + A \cdot C$ (B) $A + B \cdot C$
(C) $A \cdot B + C$ (D) $A + B$

9. How many 2-input NAND gates are required to implement a 2-input XNOR gate?

- (A) 5 (B) 4
(C) 3 (D) 6

10. What is the canonical Maxterm M2 for a 3-variable function $F(A, B, C)$?

- (A) $A + B' + C$
(C) $A' + B + C$

- (B) $A + B + C'$
(D) $A' + B' + C$

11. Simplify the expression: $F = (A + B' + C') \cdot (A + B' + C)$.

- (A) $A + B'$
(C) $B' + C$

- (B) $A + C'$
(D) $A \cdot B'$

12. What does a quad (group of 4 1s) in a 4-variable Karnaugh Map reduce to?

- (A) A product term with 2 variables
(C) A product term with 1 variable

- (B) A product term with 3 variables
(D) A single literal

13. Simplify the POS expression: $F = (P + Q) \cdot (P + Q') \cdot (P' + R)$.

- (A) $P \cdot R$
(C) $P' \cdot Q$

- (B) $P + Q \cdot R$
(D) $Q + R$

14. Identify the simplified expression for $F = A'B' + A'B + ABC + AB'C$.

- (A) $A' + C$
(C) $B + C'$

- (B) $A + B'$
(D) $A'B + C$

15. The dual of a Boolean expression is obtained by:

- (A) Swapping AND with OR, and 0 with 1 without complementing variables
(C) Swapping 0 with 1 only

- (B) Complementing all variables and operators
(D) Inverting variables only

16. Find the simplified product-of-sums (POS) for $F(A, B, C) = \prod M(1, 3, 5, 7)$.

- (A) C'
(C) $A + B$

- (B) C
(D) $A' + B'$

17. Minimize the expression: $F = (A \oplus B) \cdot A$.

- (A) $A \cdot B'$
(C) $A \cdot B$

- (B) $A' \cdot B$
(D) $A + B$

18. Simplify $F = A + A'B + A'B'C + A'B'C'D$.

- (A) $A + B + C + D$
(C) $A' + B' + C' + D'$

- (B) $A \cdot B \cdot C \cdot D$
(D) $A + B' \cdot C$

19. Which term is an Essential Prime Implicant (EPI) if it contains at least one minterm that is:

- (A) Covered by no other prime implicant
(B) Covered by all prime implicants
(C) Part of a don't-care set
(D) Located in cell 0

20. Express $F(X, Y, Z) = X + Y'Z$ in canonical minterm notation $\sum m$.

- (A) $\sum m(1, 4, 5, 6, 7)$
(B) $\sum m(0, 1, 4, 5, 7)$
(C) $\sum m(2, 3, 4, 6, 7)$
(D) $\sum m(1, 2, 3, 4, 5)$

21. Simplify $F = (A + B) \cdot (A' + C) \cdot (B + C)$ using the dual Consensus Theorem.

- (A) $(A + B) \cdot (A' + C)$
(B) $(A' + B) \cdot (A + C)$
(C) $(B + C)$
(D) $A + B + C$

22. What is the complement of $F = X'Y + XY'$ (XOR)?

- (A) $XY + X'Y'$
(B) $X'Y' + XY$
(C) Both (A) and (B)
(D) $X + Y$

23. How many entries in a 4-variable truth table yield an output of 1 for $F = A \oplus B \oplus C \oplus D$?

- (A) 8
(B) 4
(C) 12
(D) 16

24. Simplify: $F = A \cdot B + A' \cdot C + B \cdot C \cdot D$.

- (A) $A \cdot B + A' \cdot C$
(B) $A \cdot B + B \cdot C \cdot D$
(C) $A' \cdot C + B \cdot C \cdot D$
(D) $A \cdot B + A' \cdot C + B \cdot D$

25. Simplify $F = (A + B) \cdot [A' \cdot (B' + C')]'$.

- (A) $A + B$
(B) $A \cdot B$
(C) $B + C$
(D) $A + C$

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(A)	6	(B)	11	(A)	16	(A)	21	(A)
2	(A)	7	(A)	12	(A)	17	(A)	22	(C)
3	(A)	8	(A)	13	(A)	18	(A)	23	(A)
4	(A)	9	(A)	14	(A)	19	(A)	24	(A)
5	(A)	10	(A)	15	(A)	20	(A)	25	(A)